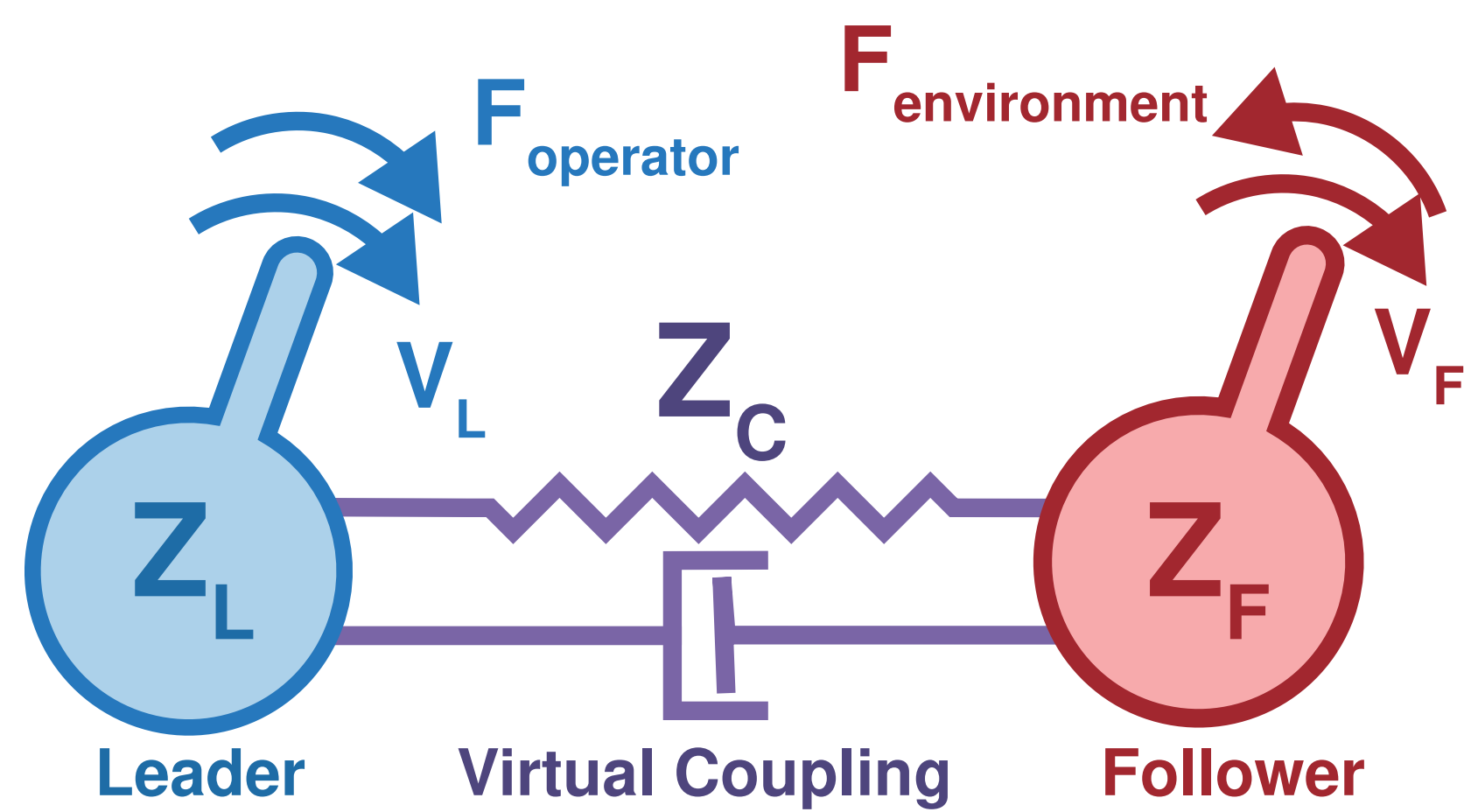
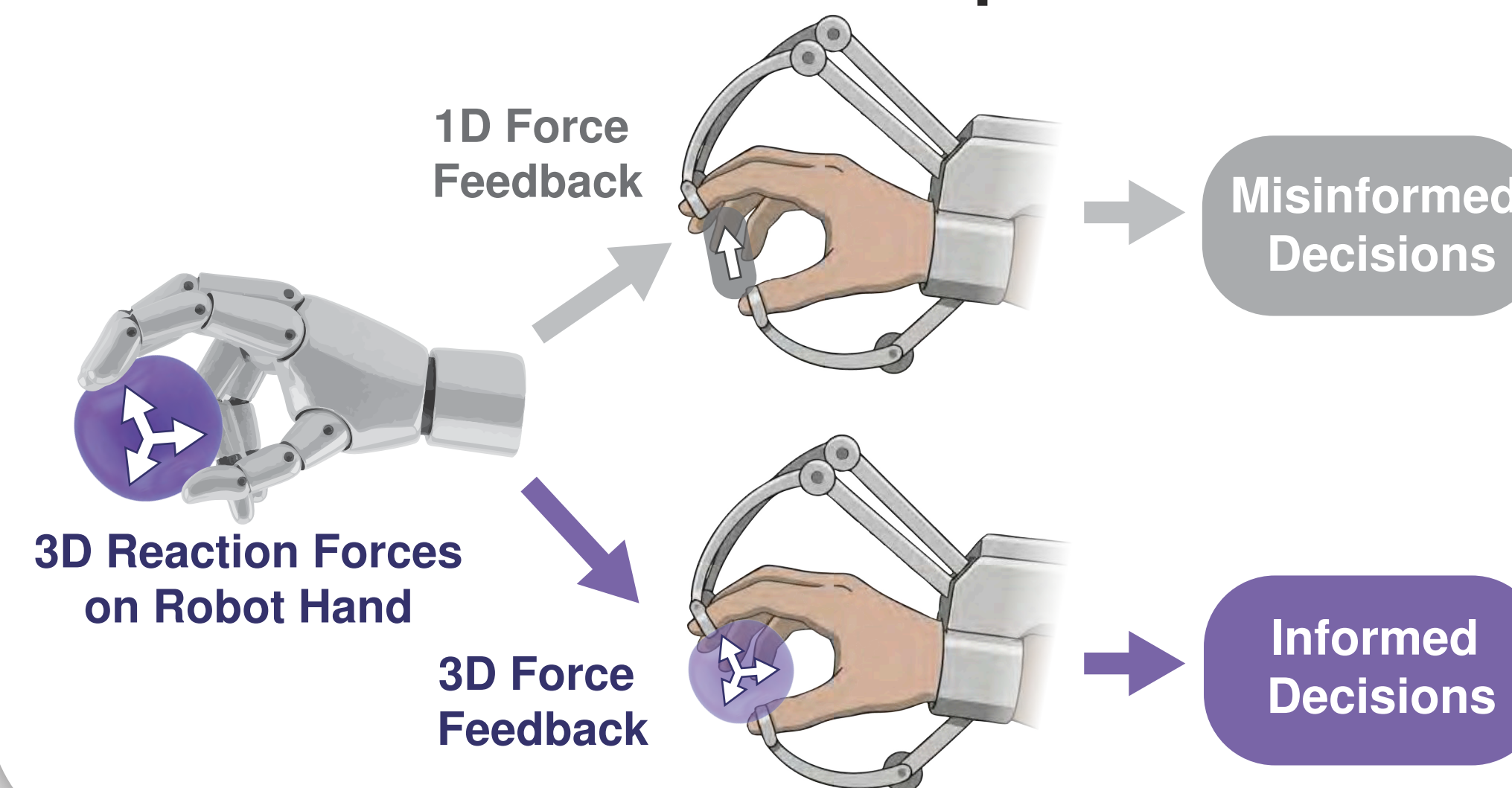


Teleoperation Performance Depends on Quality of Haptic Feedback

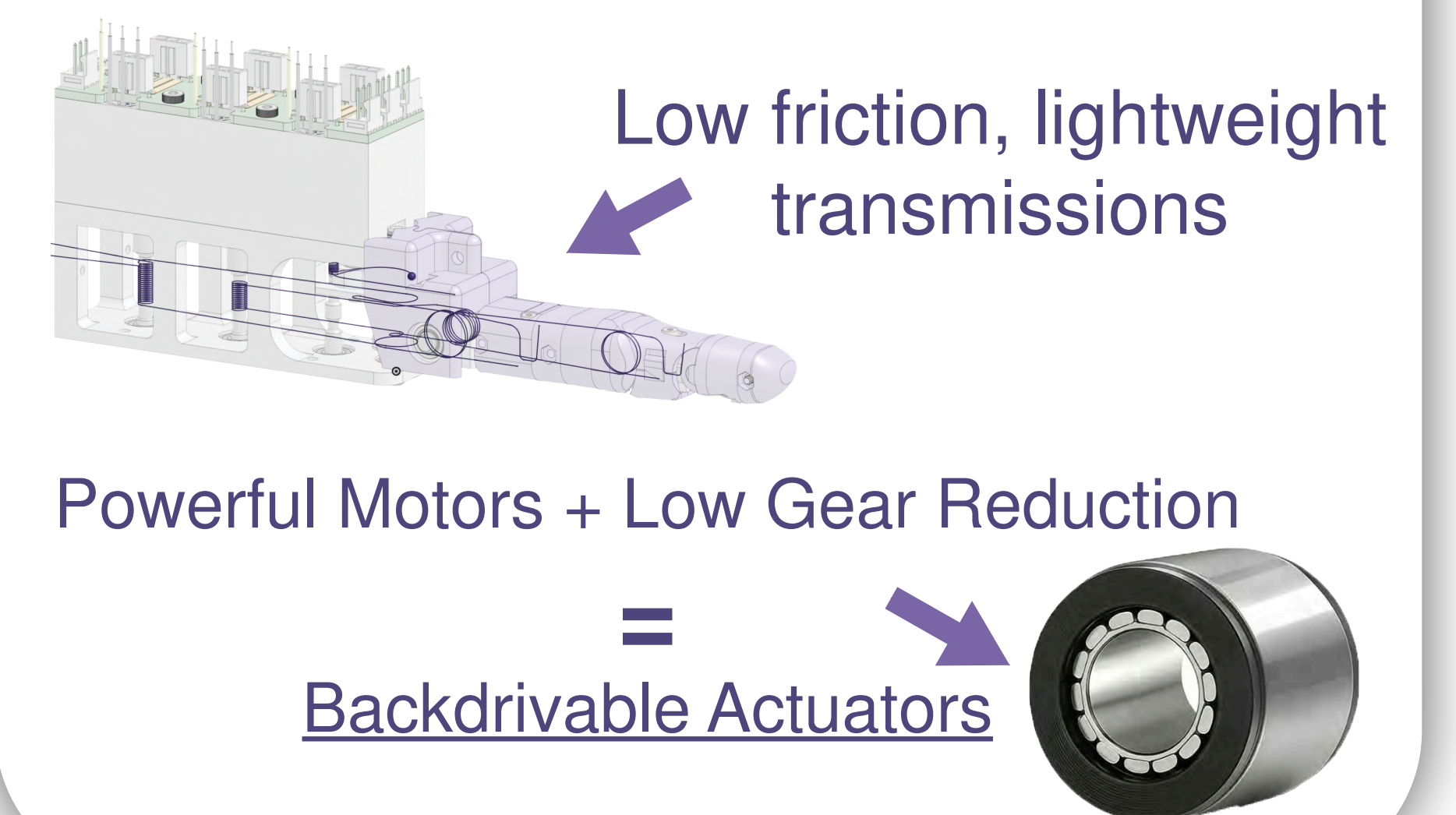
Bilateral Control Tightly Couples Leader and Follower Devices



3D Force Feedback Reveals Contact Direction and Improves Control

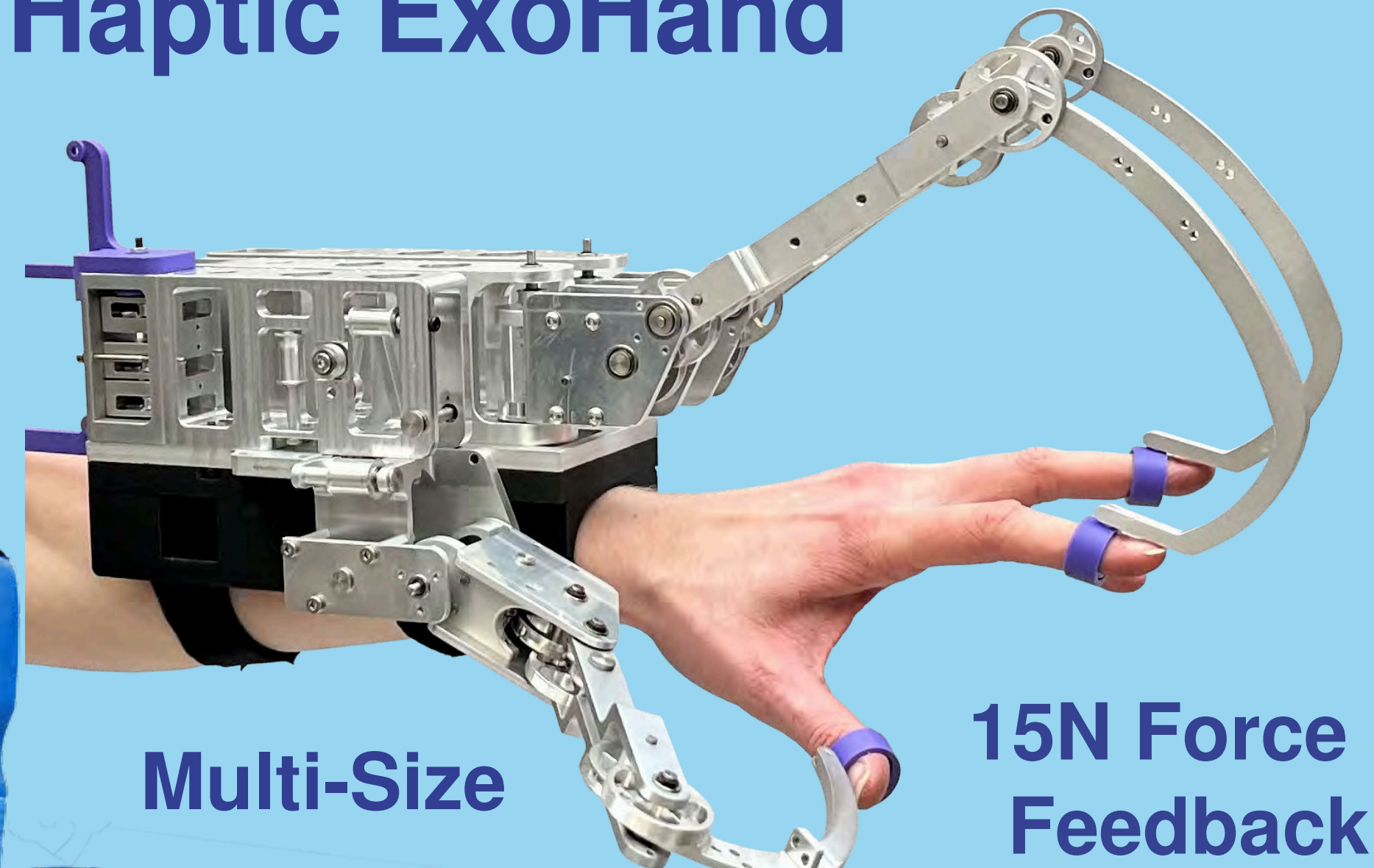


Low Mechanical Impedance Increases Force Transparency

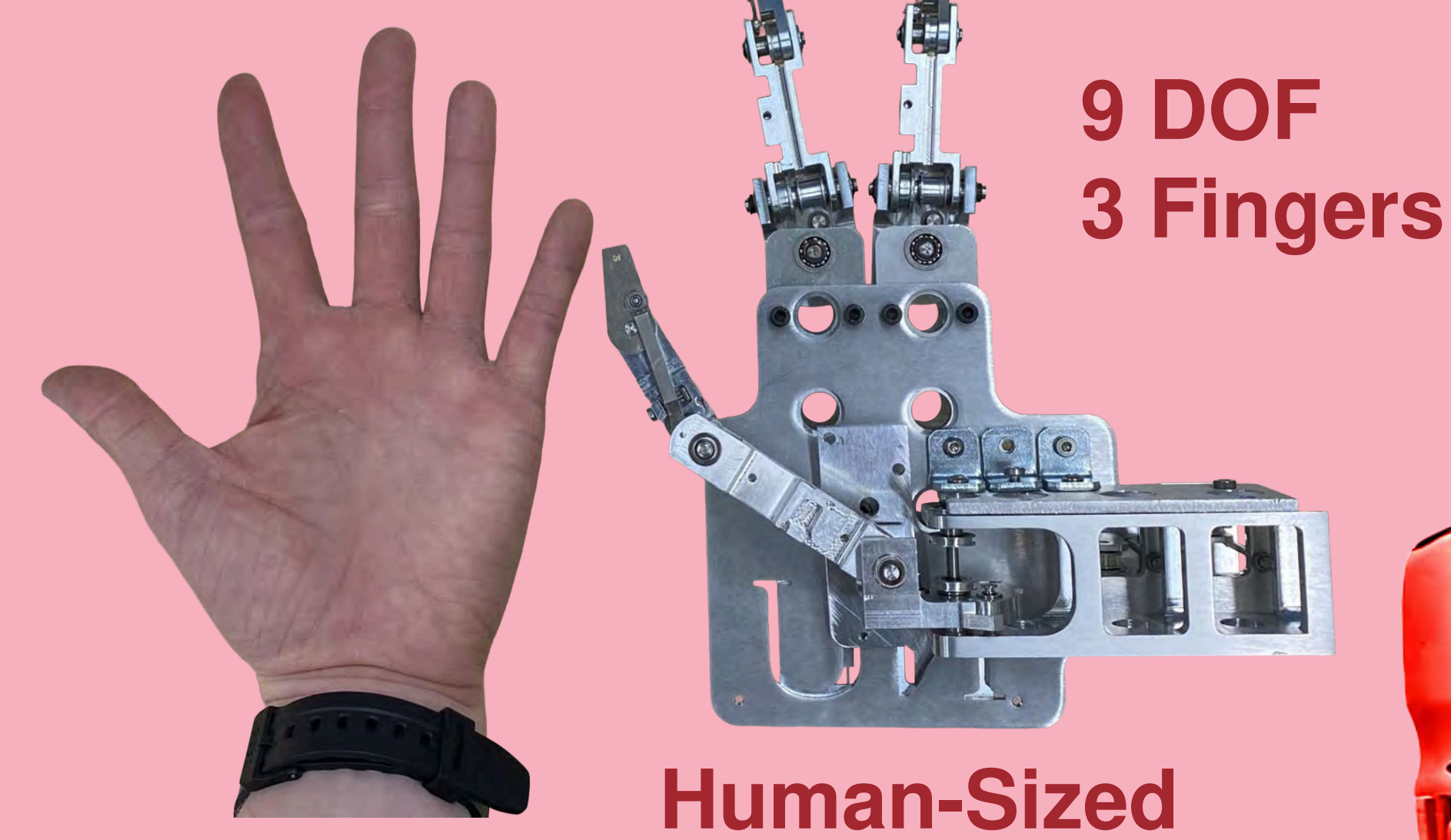


Haptic ExoHand

HAND



Robot Hand



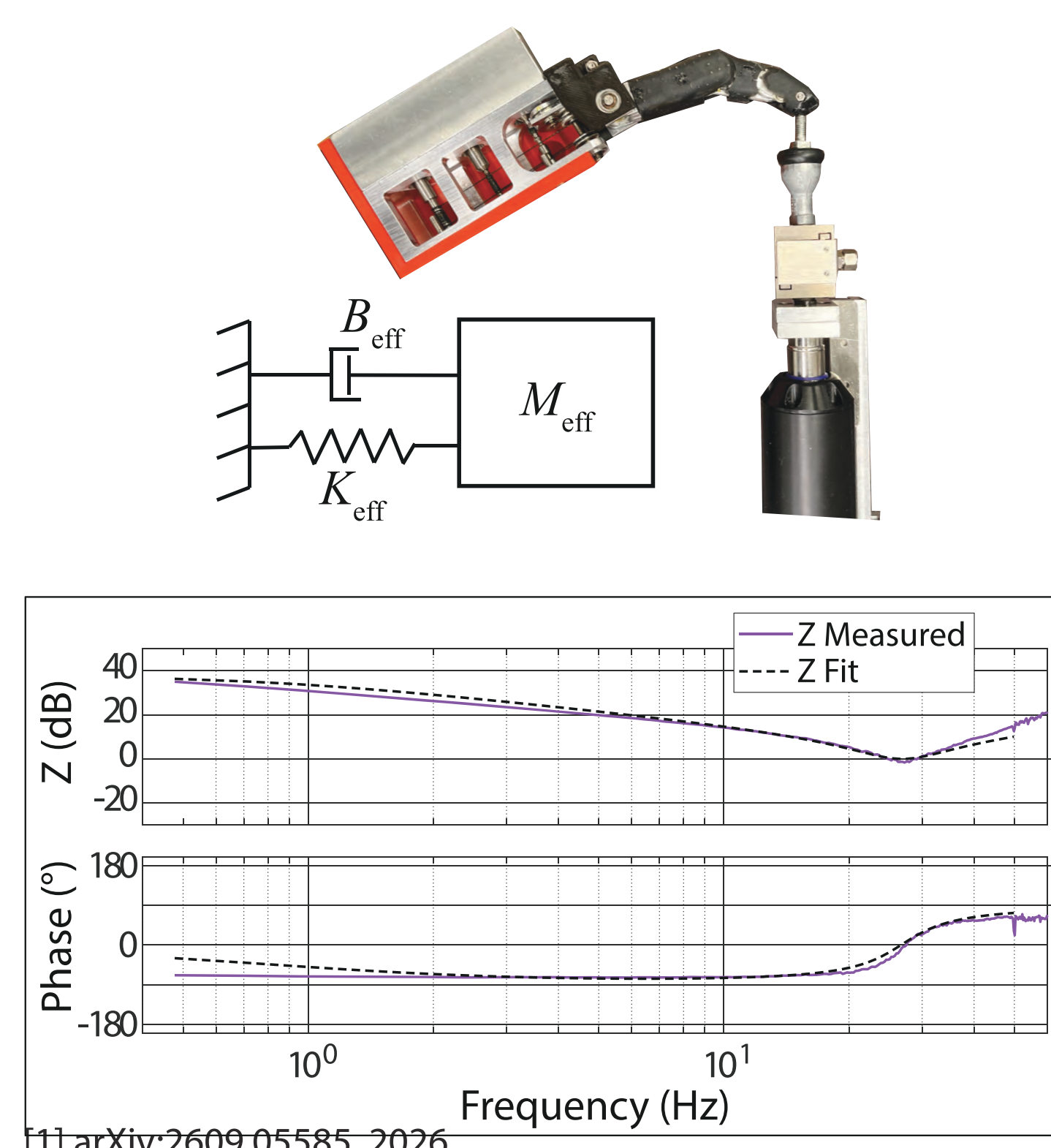
Leader

Virtual Coupling

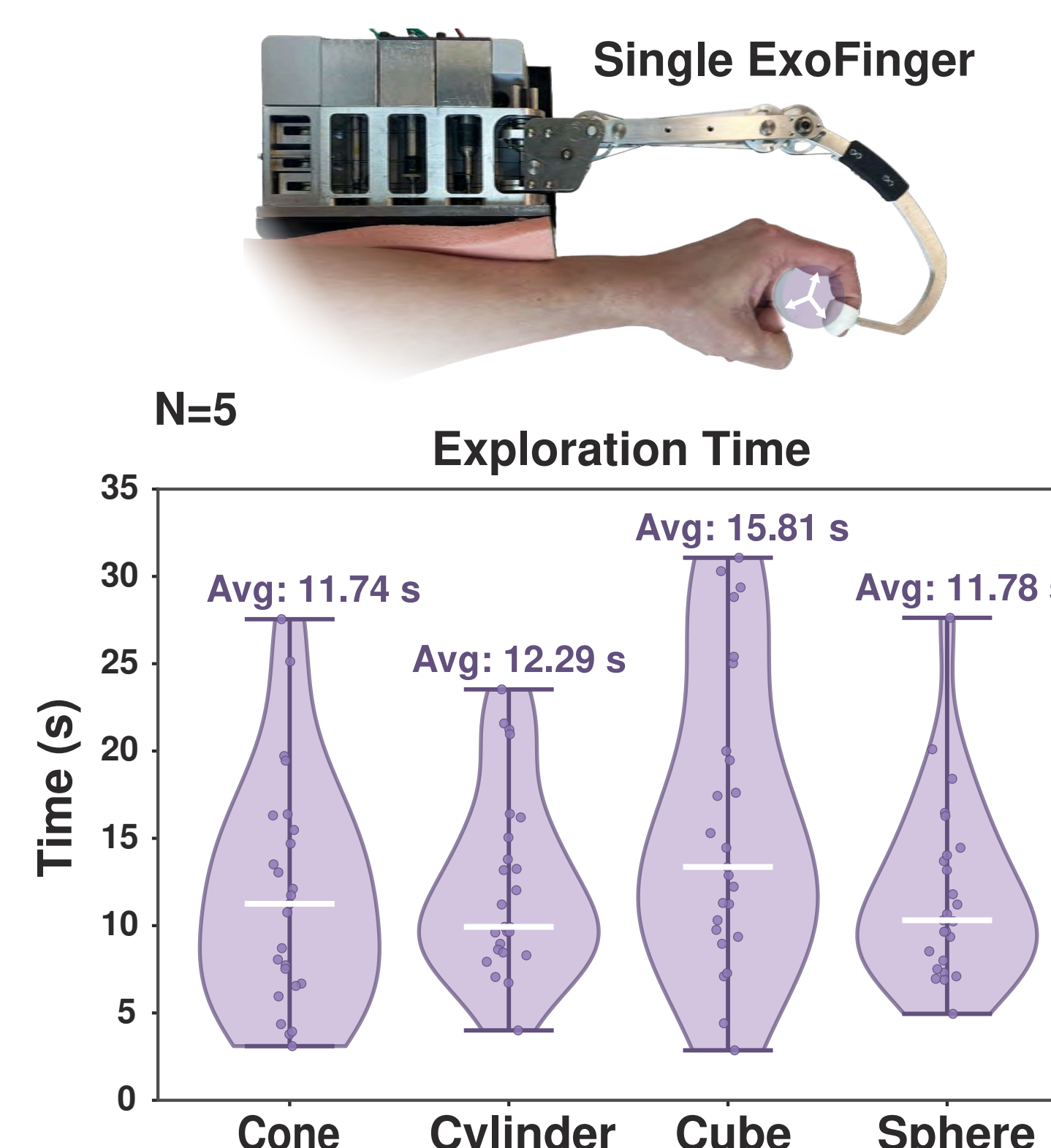
Follower

PRELIMINARY PERFORMANCE EVALUATION

Mechanical Impedance Testing



Virtual Shape Discrimination



	Cone	Cylinder	Cube	Sphere
Cone	96.0%	0.0%	0.0%	4.0%
Cylinder	4.0%	72.0%	16.0%	8.0%
Cube	8.0%	12.0%	80.0%	0.0%
Sphere	4.0%	4.0%	0.0%	92.0%